

Plan 10

Plan 10 is a series of command line tools I developed for utilizing DiffSynth Studio. DiffSynth Studio supports inferencing and training with various popular models for doing Text to Image and Text to Video tasks. The primary motivation to build this suite of tools was to make easy to use these models from command line for batch processing while also being very portable. I have this currently running on Spark DGX Arm64 and Intel i5 platforms with cuda and pytorch.

About the name

Plan 10 was named after the seminole film by Ed Wood “Plan 9 from Outer Space”. This movie is often hailed as the worst movie ever made. Ed Wood was famous for his single takes and lack of continuity between scenes. It actually reminds me a lot of current AI movies. Same background for all shots, no coherent flow between scenes, etc. I feel that I can automate these bad film decisions, hence, Plan 10 the next evolution.

Installation

To install first need cuda and anaconda installed for your system. Then copy paste the following into a command prompt to set up the conda environment, install the libraries and download loras and checkpoints.

```
git clone https://github.com/noodlepopllc/Plan10.git
cd Plan10
conda create python=3.12 -n plan10 -y
conda activate plan10
pip install -r requirements.txt
git clone https://github.com/modelscope/DiffSynth-Studio.git
pip install -e DiffSynth-Studio
hf download lightx2v/Qwen-Image-2512-Lightning Qwen-Image-2512-Lightning-8steps-V1.0-bf16.safetensors --local-dir ./loras
hf download lightx2v/Qwen-Image-Edit-2511-Lightning Qwen-Image-Edit-2511-Lightning-8steps-V1.0-bf16.safetensors --local-dir ./loras
hf download fal/Qwen-Image-Edit-2511-Multiple-Angles-LoRA qwen-image-edit-2511-multiple-angles-lora.safetensors --local-dir ./loras
hf download DeepBeepMeep/Wan2.1
loras_accelerators/Wan21_CausVid_bidirect2_T2V_1_3B_lora_rank32.safetensors --local-dir ./loras
hf download lightx2v/Wan2.1-Distill-Loras
wan2.1_i2v_lora_rank64_lightx2v_4step.safetensors --local-dir ./loras
hf download noodlepop/Wan-Series-Converted-Safetensors --local-dir ./models/DiffSynth-Studio/Wan-Series-Converted-Safetensors
python lib/config.py
```

Tools

Image Generation

lib/image_gen.py

usage: image_gen.py [-h] [-W WIDTH] [-H HEIGHT] [-E SEED] [-P PROMPT] [-O OUTPUT] [-C] [-L]

options:

-h, --help show this help message and exit
-W WIDTH, --width WIDTH width of output
-H HEIGHT, --height HEIGHT height of output
-E SEED, --seed SEED seed
-P PROMPT, --prompt PROMPT prompt
-O OUTPUT, --output OUTPUT
-C, --character-sheet
-L, --location

python lib/image_gen.py -P "anime woman in a red dress standing before a fantasy castle surrounded by orcs" -O examples/anime_red_dress.png

python lib/image_gen.py -P "a fantasy scene featuring a regal woman in an ornate red gown with gold embroidery, She wears a red flower in her hair and elegant jewelry, exuding nobility, suggesting a medieval or high-fantasy setting. storybook atmosphere. The composition blends elegance and strength, evoking themes of royalty, adventure, and fantasy lore." -O examples/char_red_dress.png -C

python lib/image_gen.py -P "This image depicts a fantasy scene featuring a majestic castle with blue-topped spires. suggesting a medieval or high-fantasy setting. The sky is bright with fluffy clouds, enhancing the epic, storybook atmosphere. The composition blends elegance and strength, evoking themes of royalty, adventure, and fantasy lore." -O examples/castle.png -L

Image Editing

usage: image_edit.py [-h] [-I IMAGES] [-P PROMPT] [-W WIDTH] [-H HEIGHT] [-E SEED] [-O OUTPUT]

options:

-h, --help show this help message and exit
-I IMAGES, --images IMAGES Input images
-P PROMPT, --prompt PROMPT Edit prompt
-W WIDTH, --width WIDTH
-H HEIGHT, --height HEIGHT
-E SEED, --seed SEED
-O OUTPUT, --output OUTPUT

python lib/image_edit.py -I examples/char_red_dress.png -P "Change the dress color to purple" -O examples/char_purple_dress.png -W 1328 -H 1328

python lib/image_edit.py -I examples/char_purple_dress.png -P "Change to closeup of the woman laughing" -O examples/char_purple_laugh.png -W 1328 -H 1328

Graphics

```
usage: graphics_gen.py [-h] [-P PROMPT] [-W WIDTH] [-H HEIGHT] [-E SEED] [-O
OUTPUT] [-V VIDEO] [-S {blur,solid,gradient_edge}]
```

ERNIE Graphic Generation + Video Frame Padding

options:

```
-h, --help                show this help message and exit
-P PROMPT, --prompt PROMPT
-W WIDTH, --width WIDTH
-H HEIGHT, --height HEIGHT
-E SEED, --seed SEED
-O OUTPUT, --output OUTPUT
-V VIDEO, --video VIDEO
                        Target video size like 1920x1080
-S {blur,solid,gradient_edge}, --padding {blur,solid,gradient_edge}
```

```
python lib/graphics_gen.py -P "A movie poster featuring anime woman in a red
dress (a fantasy scene featuring a regal woman in an ornate red gown with gold
embroidery, She wears a red flower in her hair and elegant jewelry, exuding
nobility, suggesting a medieval or high-fantasy setting.) standing before a
fantasy castle surrounded by orcs, with a title of Queen B and subcaption, orcs
don't know who they are messing with " -O examples/title_card.png -W 768 -H
1376 -V "720x1280"
```

Compositing

usage: compositor.py [-h] [-W WIDTH] [-H HEIGHT] [-E SEED] [-O OUTPUT] [-B BACKGROUND] [-C CHARS] [-S SHOT_TYPE] [-A ACTION] [-R] [-Z ZONE]

options:

- h, --help show this help message and exit
- W WIDTH, --width WIDTH
- H HEIGHT, --height HEIGHT
- E SEED, --seed SEED
- O OUTPUT, --output OUTPUT
- B BACKGROUND, --background BACKGROUND
Background path
- C CHARS, --chars CHARS
Character paths (1-2)
- S SHOT_TYPE, --shot-type SHOT_TYPE
- A ACTION, --action ACTION
Action to complete
- R, --gen-reverse Generate reverse-angle background (T2I)
- Z ZONE, --zone ZONE Request different zone other than reverse

```
python lib/image_gen.py -L -P "Interior of a fantasy castle throne room" -O
examples/throne.png
```

```
python lib/image_gen.py -C -P "The queen's personal lady in waiting eloquent,
fantasy medieval style, (23 year old woman, plain tight fitting dress, blonde
hair in pigtails)" -O examples/lady_waiting.png
```

```
python lib/compositor.py -B examples/throne.png -C examples/char_red_dress.png
-C examples/lady_waiting.png -S "two_shot" -A "The woman in the red dress sits
upon her throne (gaze towards), the 24 year old woman in green dress and
pigtails kneels before her with head bowed respectfully" -O
examples/throne_sitting.png
```

```
python lib/compositor.py -B examples/throne.png -C examples/char_red_dress.png
-S "closeup" -A "The woman in the red dress sits upon her throne (looking
forward with evil grin)" -O examples/throne_closeup.png
```

```
python lib/compositor.py -B examples/throne.png -R -Z "large double wooden doors
with decorative torches on either side" -O examples/throne_doors.png
```

```
python lib/compositor.py -B examples/throne_doors.png -C
examples/lady_waiting.png -S "medium" -A "The woman in the green dress with
pigtails stands in front of the doors, hands in front, head bowed respectfully"
-O examples/lady_doors.png
```

```
python lib/compositor.py -B examples/throne.png -C examples/lady_waiting.png -C
examples/char_red_dress.png -S "ots" -A "The woman in red is sitting on the
throne looking up annoyed" -O examples/lady_ots.png
```

